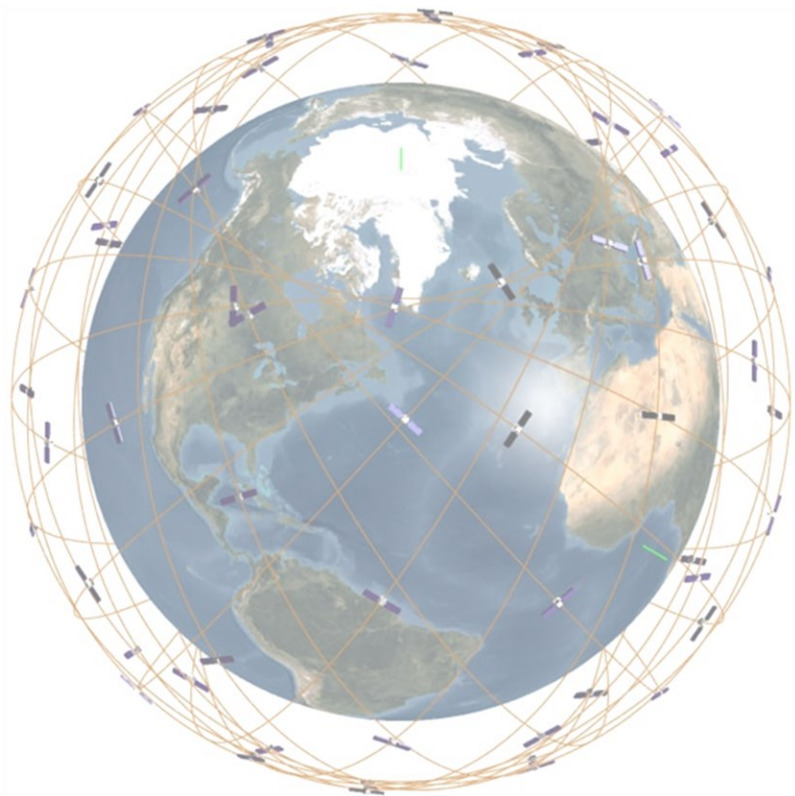






Cybersecurity for Space System (CSS) Simulator

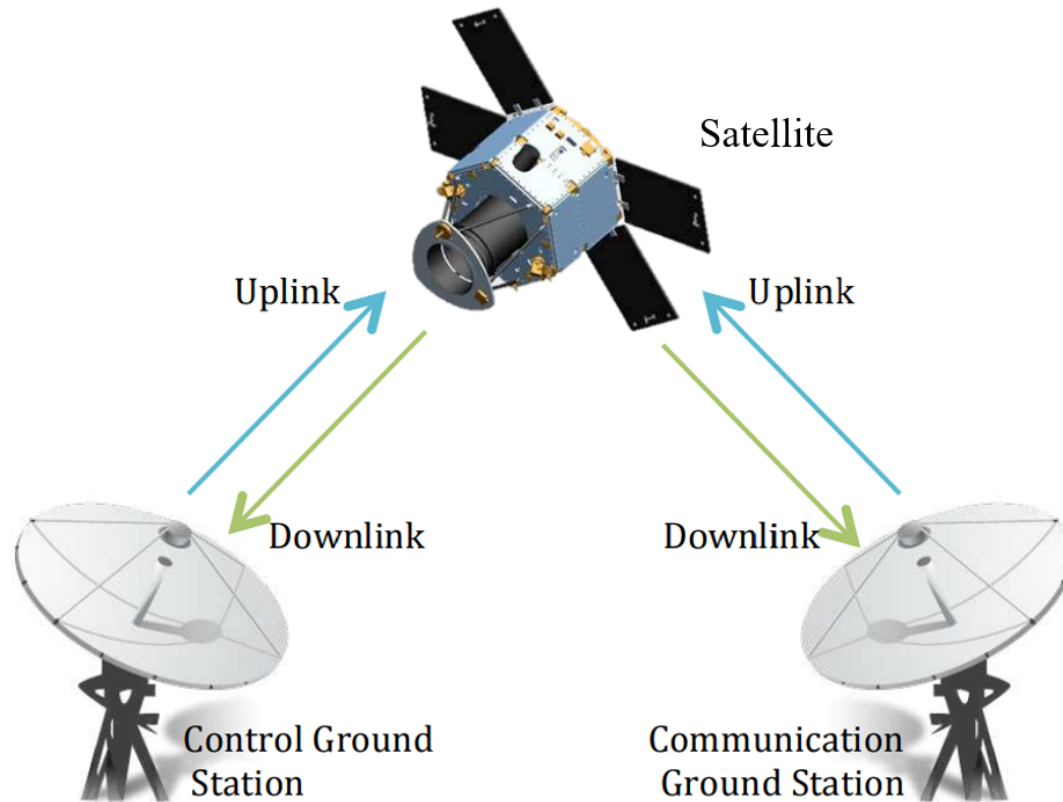
The simulator is developed in the framework of CSS project in Toulouse.

The simulator is based on NASA NOS3 v1.6.2. (released 07/08/2023) :
https://github.com/nasa/nos3/releases/tag/v1_06_02

The reference mission is STF-1: <https://www.nasa.gov/simulation-to-flight-1/>

What are the hypotheses ?

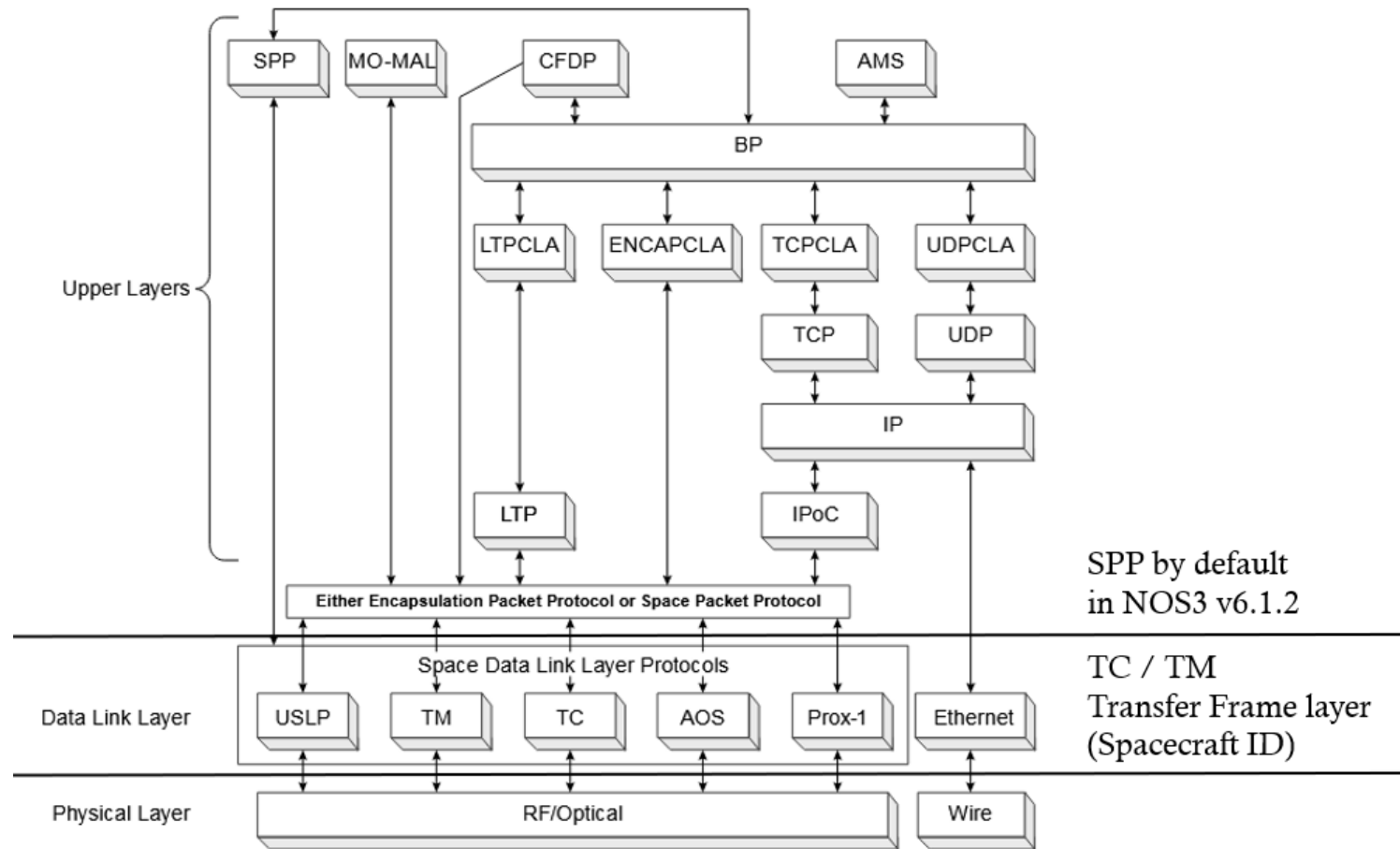
HP : CCSDS Stack over UDP = no RF !



- We can use tcpdump
- Easier attack – defence testing
- Lighter simulation (no RF)

(But missing attacks related to RF part)

What is CCSDS ?



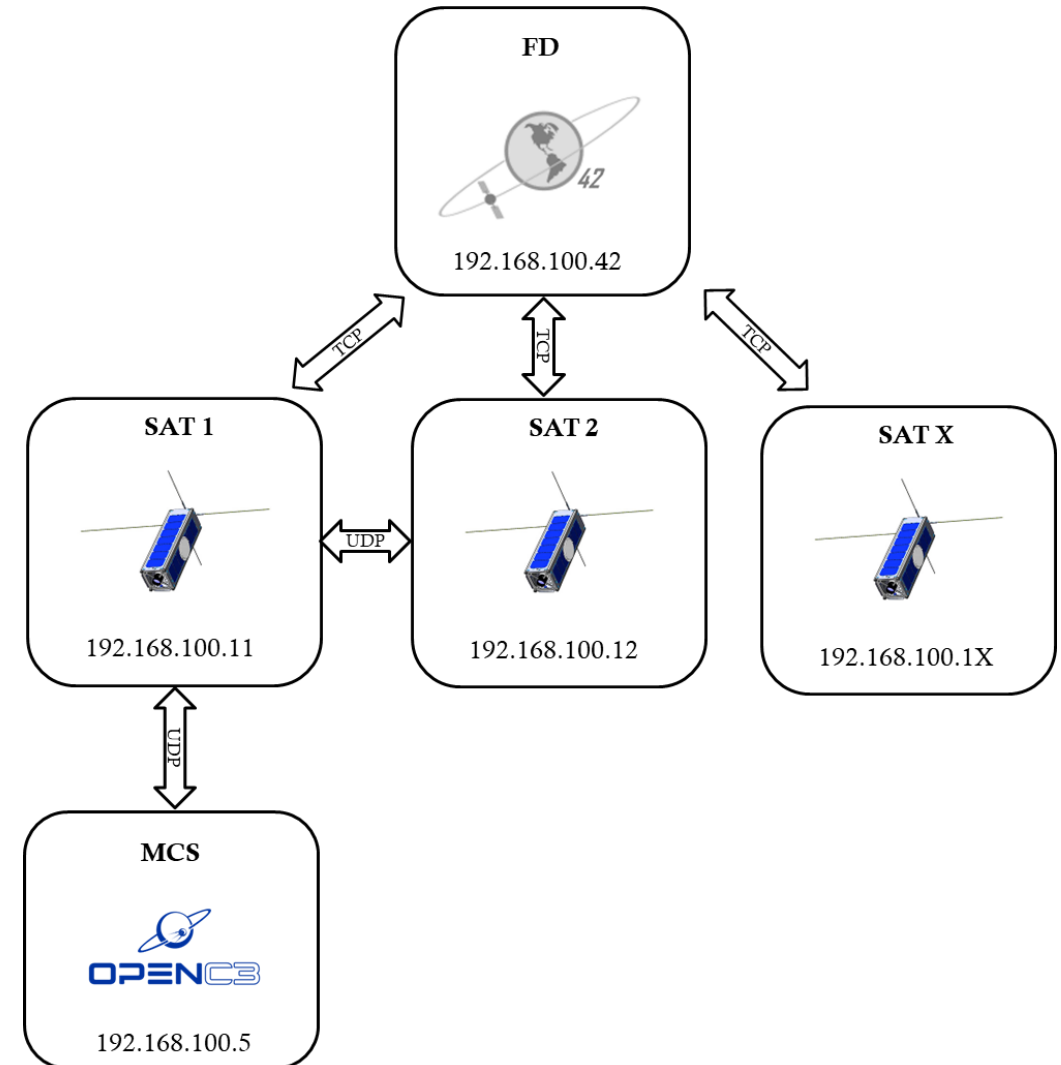
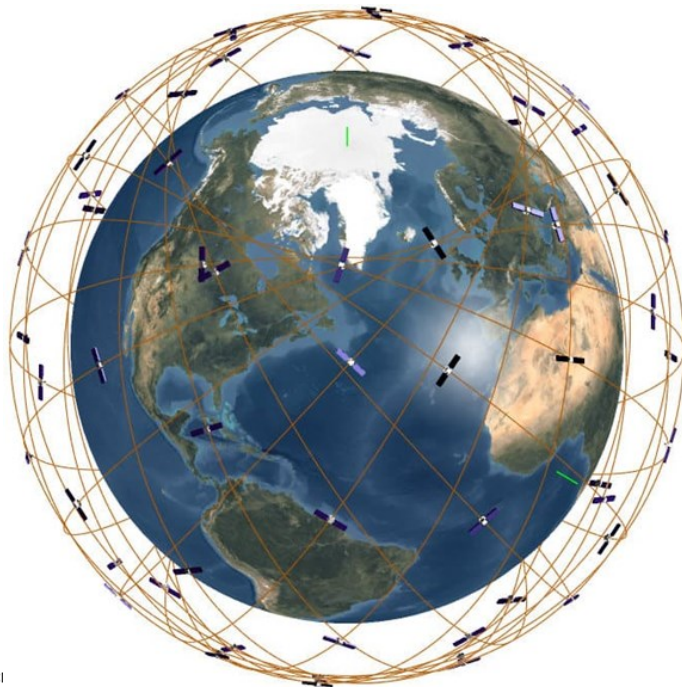
CCSDS 132.0-B-3
232.0-B-4

Figure 2-1: SPP Context within the CCSDS Protocol Stack

How CSS Simulator works ?

CSS Simulator = Multiple VMs in CITEF
that can be deployed based on a single asset.
The main asset is based on NASA NOS3.

→ We can create a constellation of CubeSats
+ Mission Control System (MCS) + Flight Dynamics



What is NASA NOS3 ?

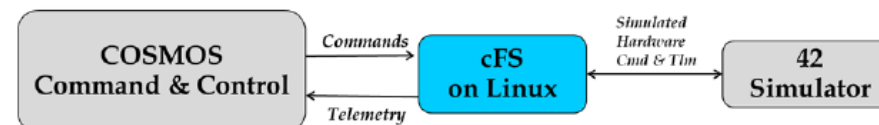
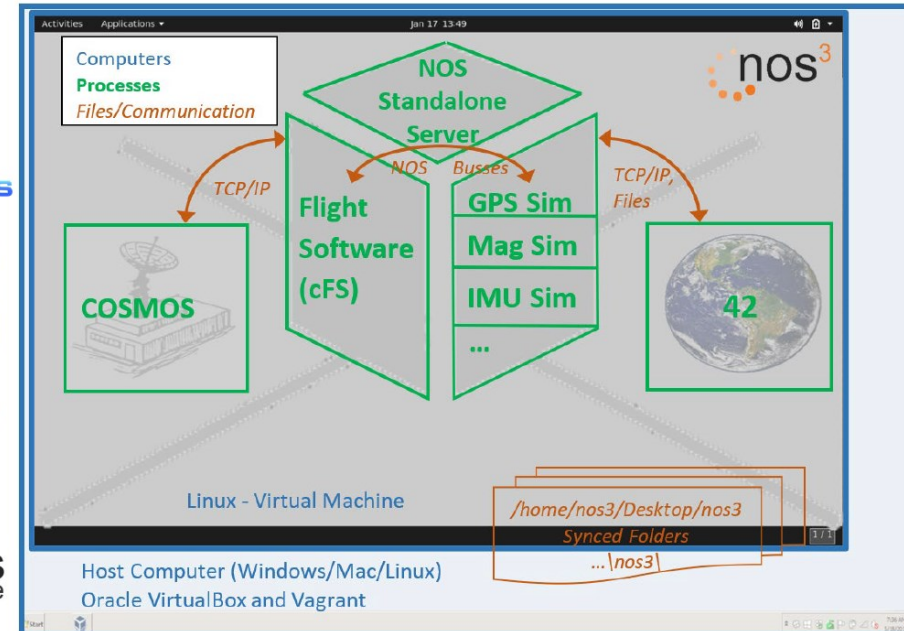
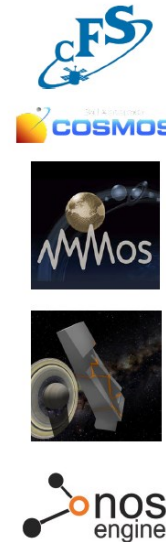
→ CSS Simulator is based on NASA NOS3.



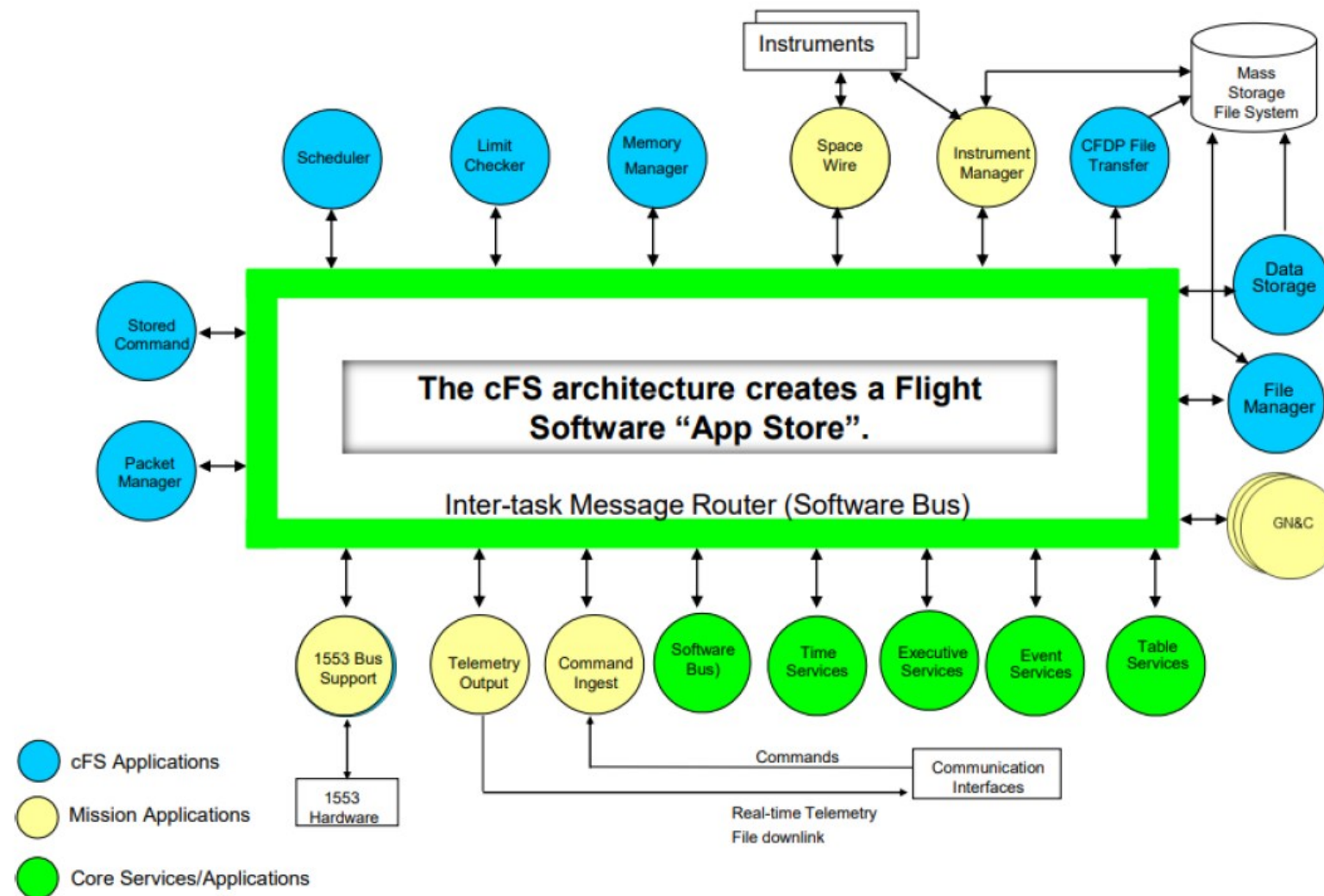
NOS3 Architecture



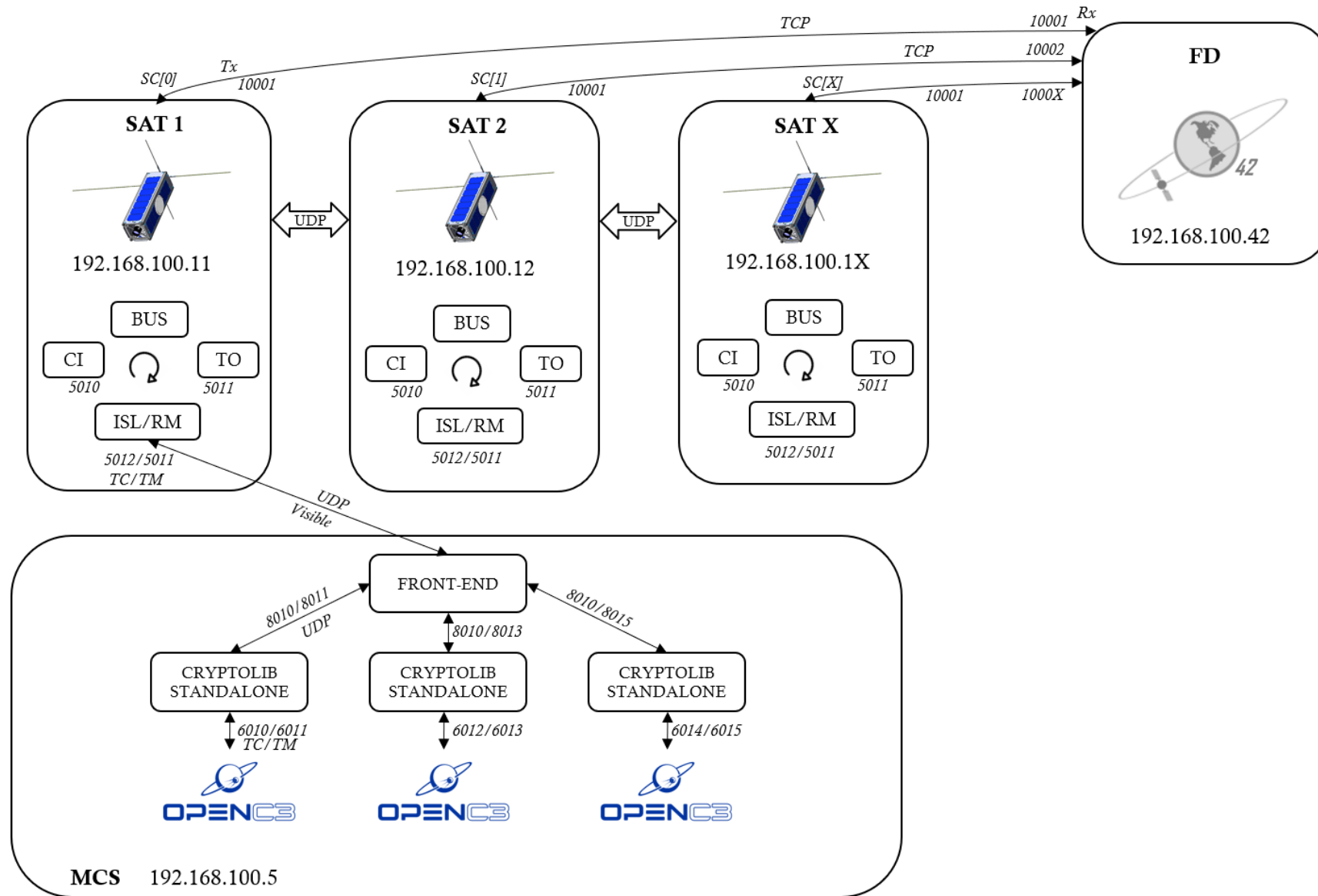
- Combination of multiple pieces of open-source software
 - A virtual spacecraft!
 - Learn on, do research, baseline as your own
- Flight Software
 - Core Flight System (cFS)
- Ground Stations
 - Ball Aerospace COSMOS
 - JPL AMMOS Instrument Toolkit
 - OpenC3 COSMOS
- Dynamics / Visualization
 - 42
- NOS Engine
 - Communication protocols
 - Hardware abstraction layer
 - Component simulators



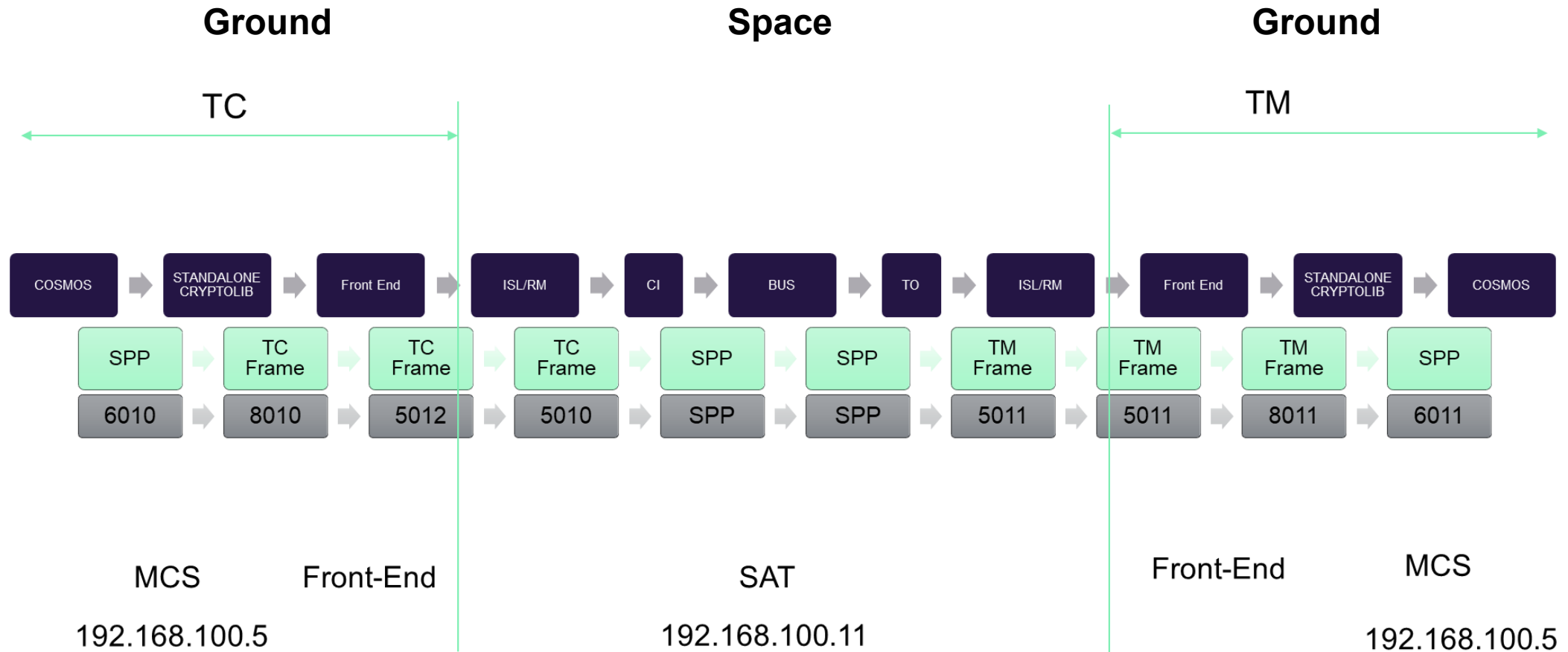
What is NASA Flight Software (cFS) ?



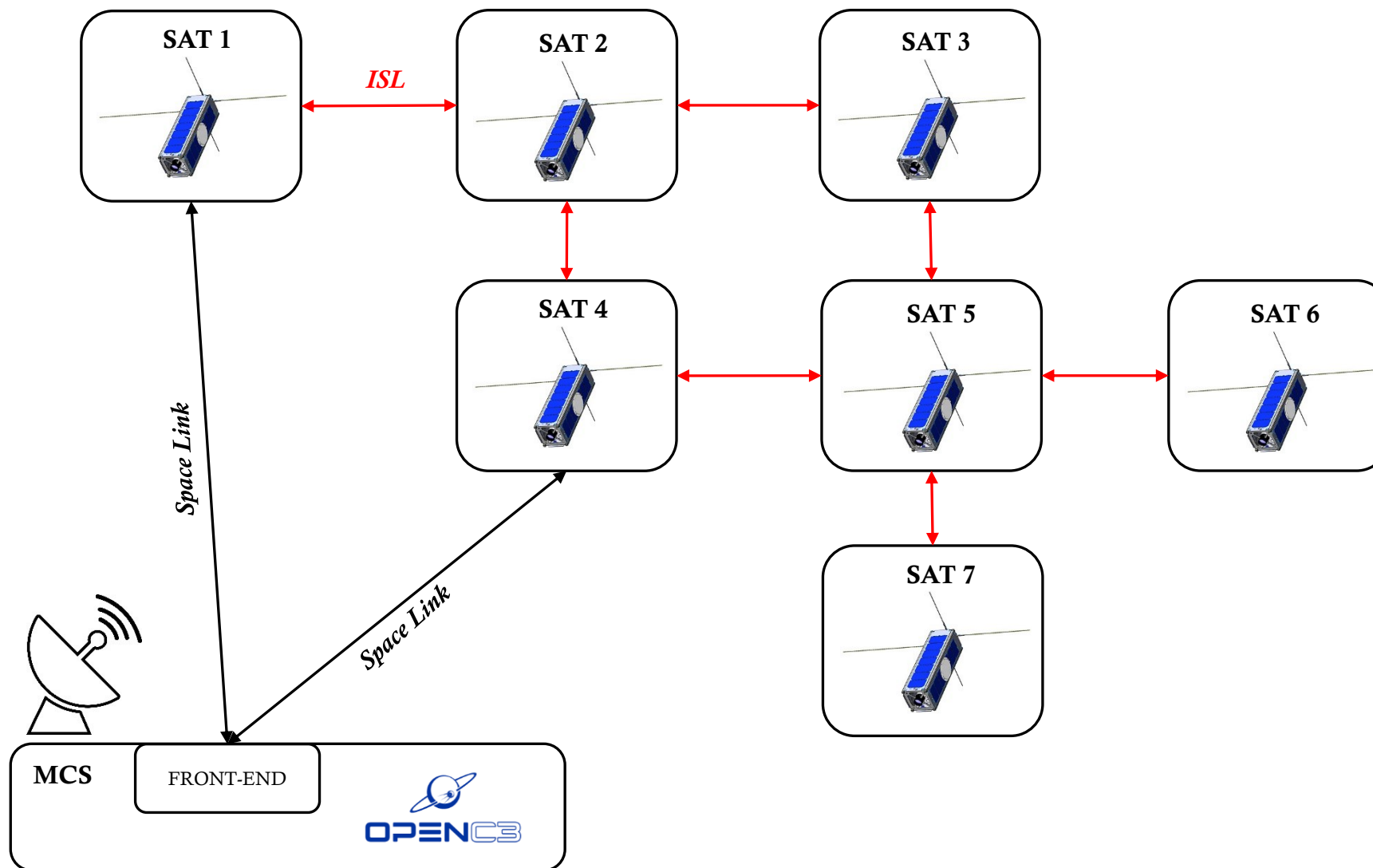
What is CSS Simulator Architecture ?



Example of TC / TM traffic (Space Link) ?



How is TC / TM Routing simulated ?



How is TC / TM Routing simulated ?

Constellation Route Configuration (text file)

Dest ID	0	1	2	3	4	5	6	7
Node ID 0	0	1	1	1	4	4	4	4
Node ID 1	0	1	2	2	2	2	2	2
Node ID 2	1	1	2	3	4	3	3	3
Node ID 3	2	2	2	3	5	5	5	5
Node ID 4	0	2	2	2	4	5	5	5
Node ID 5	4	4	4	3	4	5	6	7
Node ID 6	5	5	5	5	5	5	6	5
Node ID 7	5	5	5	5	5	5	5	7

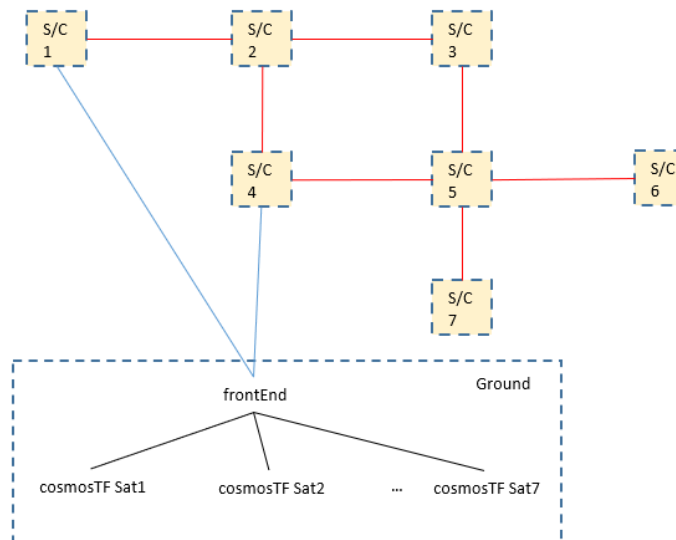
Destinations

Line for Front End
routing config.
(node 0)

Other lines for
ISL components
of sats (S/C)

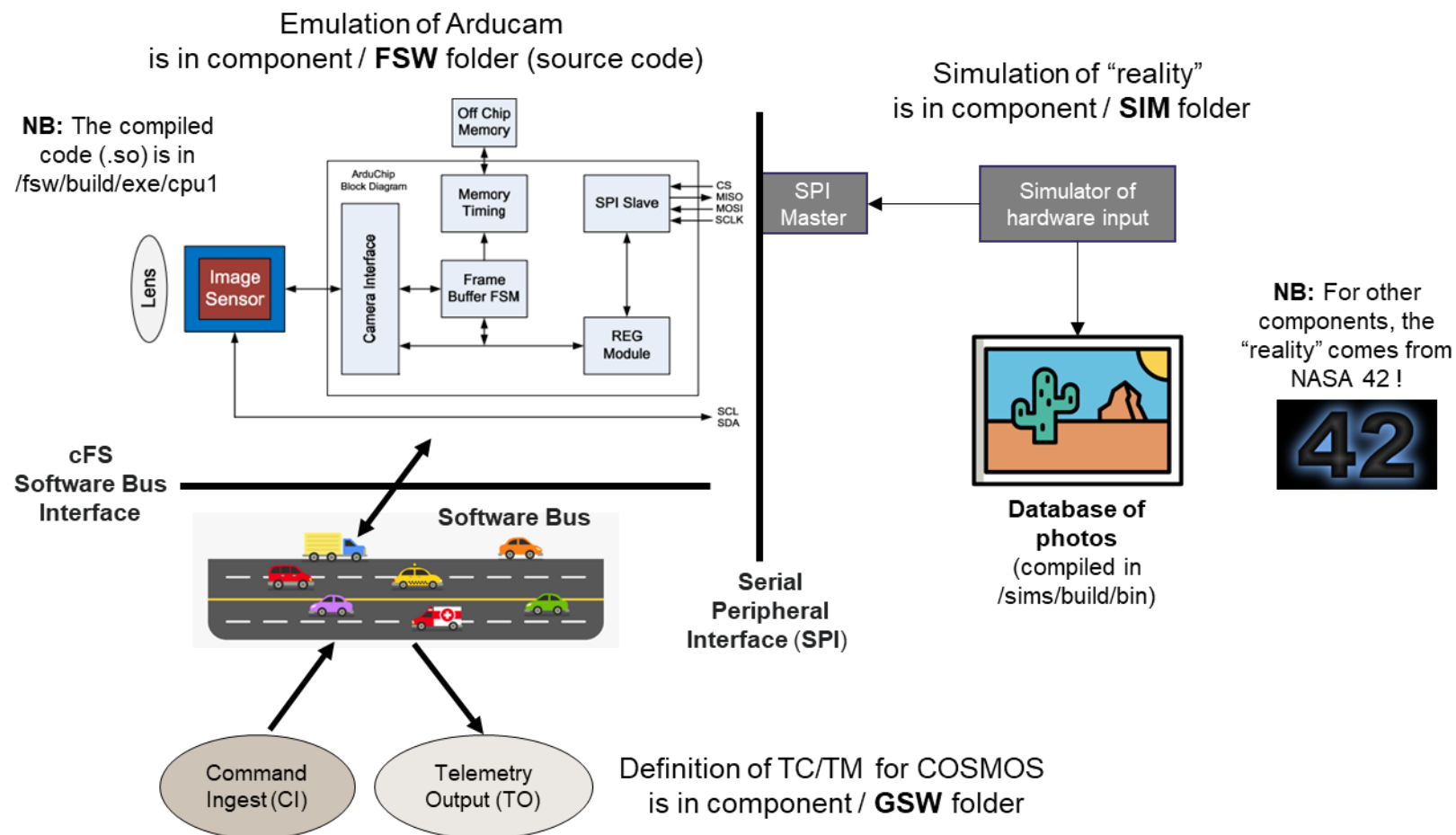
Node 0 is Front End (ground)
Node i is satellite i

In this **exemple** only S/C1 and
S/C4 are in visibility



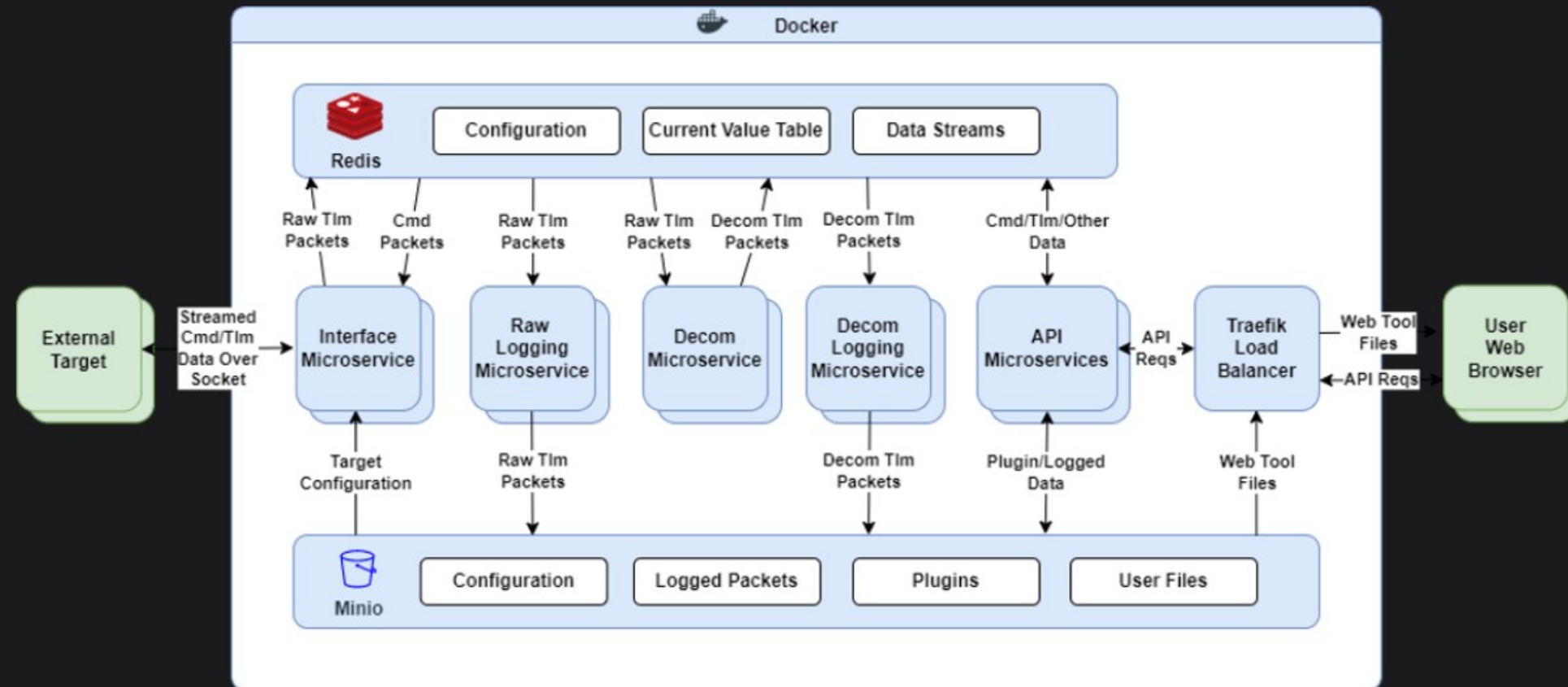
How are Flight Software, Ground Software and Environment simulated for the satellite components ?

Arducam Example



What is COSMOS ?

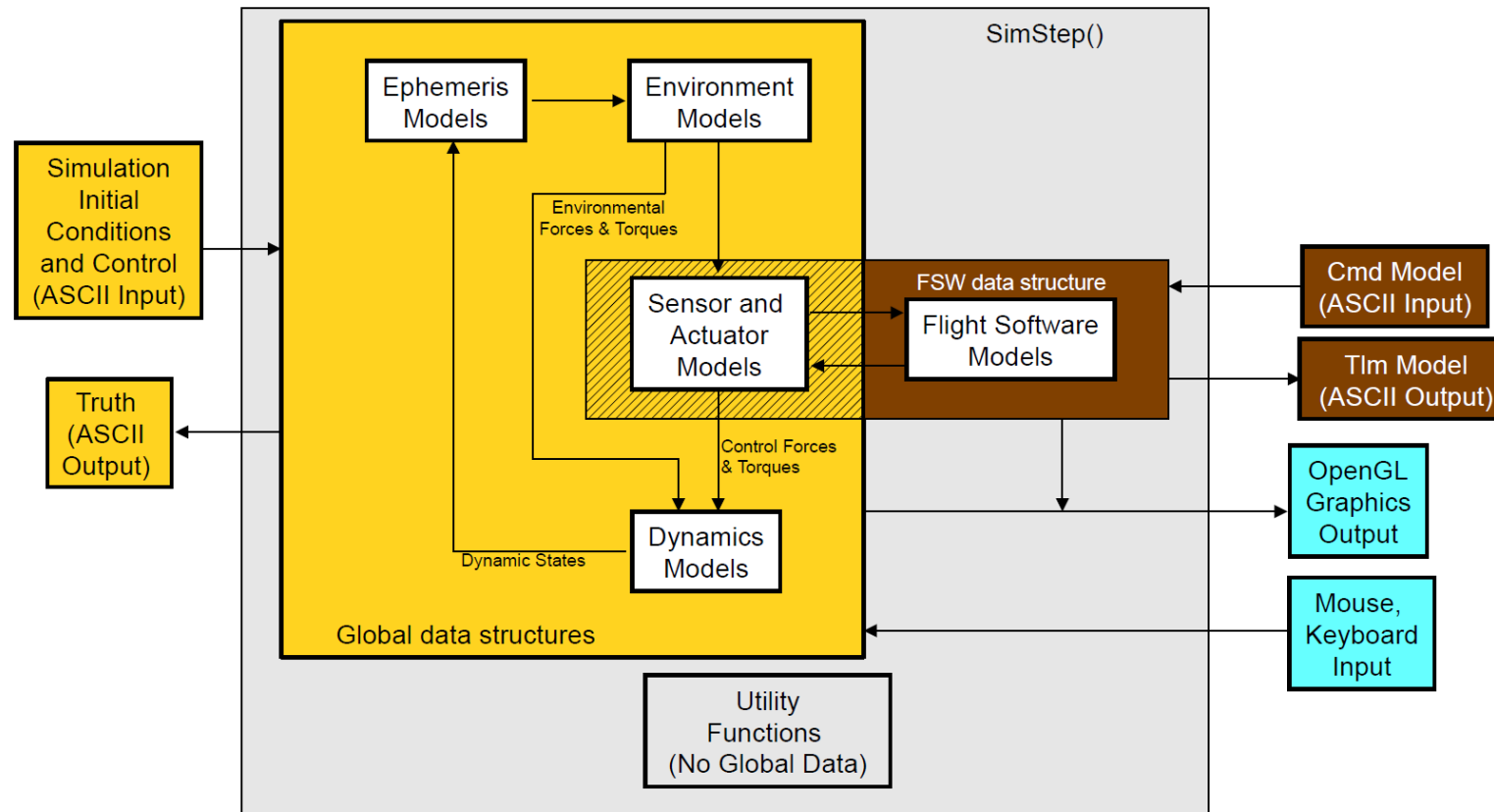
The following diagram shows the COSMOS 5 architecture.



* Not all microservices and data flows shown

What is NASA 42 ?

42's Software Architecture



<https://github.com/ericstoneking/42>

How to start ?

Quick start:

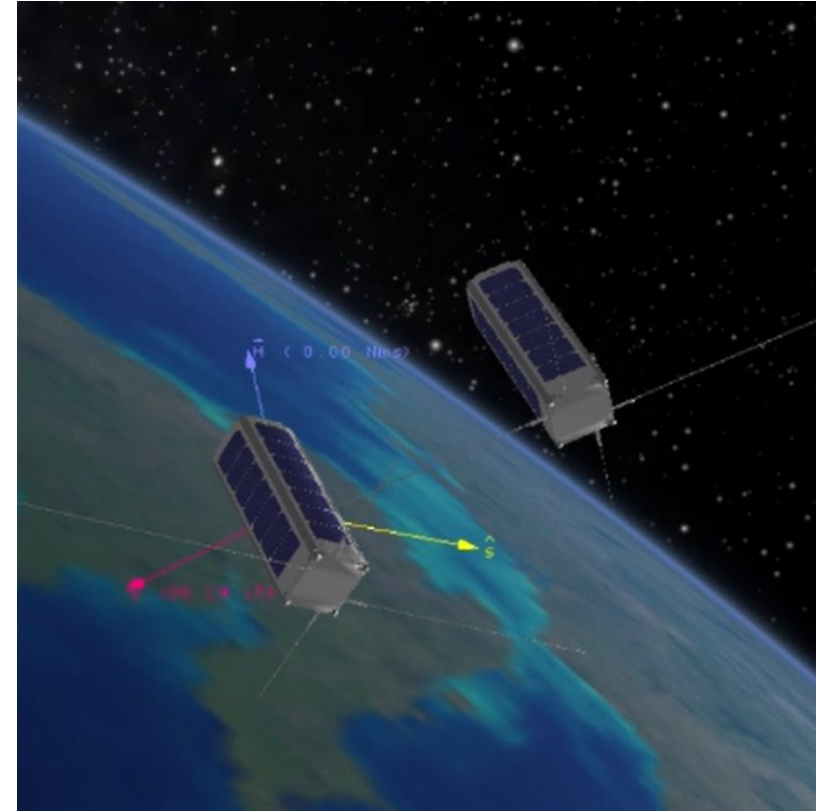
- To launch the simulator, you can just log to SAT(.11) and launch `./start.sh` in github-nos3 folder. (to stop you can use `./stop.sh`)

After launch:

- NOS3 Flight software tabs will be available in VM SAT(.11)
- COSMOS GUI for TM/TC management will be available on VM MCS(.5)
- Satellites visualization will be available in VM VISU(.42)

Attacks:

- You can test some exploits in CSS_Attacks folder by launching `./Master_Attacks_Scripts.sh` + the number of the exploit you want to test.



Back-up slides



